



CSE3023 Web-Based Application Development

Semester II 2025/26

Lab Report 7

Group:K1

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Lab Exercise

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>Register Subject</title>
</head>
<body>
    <h2>Register New Subject</h2>
    <form action="../SubjectServlet" method="POST">
        <input type="hidden" name="action" value="add" />
        Subject Code: <input type="text" name="subjectCode" required /><br><br>
        Subject Name: <input type="text" name="subjectName" required /><br><br>

        <input type="submit" value="Register" />
        <a href="../dashboard.jsp">Cancel and Back</a>
    </form>
</body>
</html>
```

registerSubject.jsp

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="com.lab.bean.SubjectBean" %>
<!DOCTYPE html>
<html>
<head>
    <title>Edit Subject</title>
</head>
<body>
    <h2>Edit Subject Details</h2>
    <%
        SubjectBean sub = (SubjectBean) request.getAttribute("subject");
        if (sub != null) {
    <%
        <form action="SubjectServlet" method="POST">
            <input type="hidden" name="action" value="update" />
            <input type="hidden" name="id" value="<%= sub.getId() %>" />

            Subject Code: <input type="text" name="subjectCode" value="<%= sub.getSubjectCode() %>" required /><br><br>
            Subject Name: <input type="text" name="subjectName" value="<%= sub.getSubjectName() %>" required /><br><br>
            <input type="submit" value="Save Changes" />
            <a href="SubjectServlet?action=view">Cancel</a>
        </form>
    <%
        } else {
            out.println("<p style='color:red;'>Subject not found.</p>");
        }
    <%
    </body>
</html>
```

updateSubject.jsp

```

<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="java.util.ArrayList" %>
<%@ page import="com.lab.bean.SubjectBean" %>
<!DOCTYPE html>
<html>
<head>
<title>My Subjects</title>
</head>
<body>
<h2>Registered Academic Subjects</h2>
<p><a href="subject/registerSubject.jsp">+ Register New Subject</a> | <a href="dashboard.jsp">🏠 Dashboard</a></p>

<table border="1" cellpadding="8" cellspacing="0">
<tr style="background-color: #f2f2f2;">
<th>No</th>
<th>Subject Code</th>
<th>Subject Name</th>
<th>Actions</th>
</tr>
<%
    ArrayList<SubjectBean> list = (ArrayList<SubjectBean>) request.getAttribute("subjectList");
    if (list == null || list.isEmpty()) {
<%
        <tr>
            <td colspan="4">No subjects found.</td>
        </tr>
<%
    } else {
        for (int i = 0; i < list.size(); i++) {
            SubjectBean sub = list.get(i);
<%
            <tr>
                <td><%= i + 1 %></td>
                <td><%= sub.getSubjectCode() %></td>
                <td><%= sub.getSubjectName() %></td>
                <td>
                    <a href="SubjectServlet?action=edit&id=<%= sub.getId() %>" class="btn btn-sm btn-warning">Edit</a> ||
                    <a href="SubjectServlet?action=delete&id=<%= sub.getId() %>"
                        onclick="return confirm('Adakah anda pasti mahu memadam subjek ini? (Are you sure you want to delete this subject?);'"
                        class="btn btn-sm btn-danger">Delete
                    </a>||
                    <a href="subject/registerSubject.jsp">Register New Subject</a> ||
                    <a href="dashboard.jsp">Back to Dashboard</a>
                </td>
            </tr>
<%
        }
    }
<%>
</table>
</body>
</html>

```

viewSubject.jsp

```

<%%page contentType="text/html" pageEncoding="UTF-8"%>
<%%page import="com.lab.bean.StudentBean"%>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Student Dashboard</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<style>
    .profile-img {
        width: 160px;
        height: 160px;
        object-fit: cover;
        border: 5px solid #198754;
    }
</style>
</head>
<body class="bg-light">
<%
    // 1. Session Verification (Security Check)
    response.setContentType("text/html;charset=UTF-8");
    StudentBean user = (StudentBean) session.getAttribute("loggedUser");

    // 如果没有登录，强行访问直接踢回登录页
    if (user == null) {
        response.sendRedirect("login.html");
        return;
    }
%>

<div class="container d-flex justify-content-center align-items-center" style="min-height: 100vh; padding: 20px 0;">
<div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center" style="max-width: 600px; width: 100%;">
<div class="card-body">


<h2 class="fw-bold text-dark mb-2">Welcome, <%= user.getFullname() %>!</h2>
<p class="text-muted fs-5 mb-4">Matric Number: <strong class="text-secondary"><%= user.getMatricNo() %></strong></p>

<div class="card bg-white border rounded-3 p-3 mb-4 text-start">
<h5 class="fw-bold text-success mb-3">Subject Management</h5>
<div class="d-grid gap-2">
<a href="SubjectServlet?action=view" class="btn btn-outline-success fw-semibold">
    View My Registered Subjects
</a>
<a href="subject/registerSubject.jsp" class="btn btn-outline-primary fw-semibold">
    Register New Subject
</a>
</div>
</div>

<div class="d-flex justify-content-center gap-2 flex-wrap">
<a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2 fw-semibold">Logout</a>

<a href="UserServlet?action=delete"
    onclick="return confirm('WARNING: Are you sure you want to delete your account permanently?');"
    class="btn btn-danger px-4 py-2 fw-semibold">Delete Account</a>
</div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Dashboard.jsp

```
...A] login.html [-/A] x dashboard.jsp [-/A] x SubjectBean.java [-/A] x SubjectDAO.java [-/A] x SubjectServlet.java [-/A] x viewSubjects.jsp [-/A] x registerSul

Source History

3 import com.lab.bean.StudentBean;
4 import com.lab.bean.SubjectBean;
5 import com.lab.dao.SubjectDAO;
6 import java.io.IOException;
7 import java.util.ArrayList;
8 import javax.servlet.ServletException;
9 import javax.servlet.annotation.WebServlet;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13 import javax.servlet.http.HttpSession;
14
15
16 @WebServlet("/SubjectServlet")
17 public class SubjectServlet extends HttpServlet {
18     private SubjectDAO subjectDAO = new SubjectDAO();
19
20     @Override
21     protected void doGet(HttpServletRequest request, HttpServletResponse response)
22         throws ServletException, IOException {
23         doPost(request, response);
24     }
25
26     @Override
27     protected void doPost(HttpServletRequest request, HttpServletResponse response)
28         throws ServletException, IOException {
29
30
31         HttpSession session = request.getSession(false);
32         if (session == null || session.getAttribute("loggedUser") == null) {
33             response.sendRedirect("login.html");
34             return;
35         }
36
37         StudentBean student = (StudentBean) session.getAttribute("loggedUser");
38         String matricNo = student.getMatricNo();
39
40         String action = request.getParameter("action");
41         if (action == null) action = "view";
42
43         // ===== 1. READ =====
44         if ("view".equals(action)) {
45             ArrayList<SubjectBean> list = subjectDAO.getSubjectsByStudent(matricNo);
46             request.setAttribute("subjectList", list);
47             request.getRequestDispatcher("subject/viewSubjects.jsp").forward(request, response);
48         }
49
50         // ===== 2. CREATE =====
51
52 }
```

```

// ===== 2. CREATE =====
else if ("add".equals(action)) {
    SubjectBean sub = new SubjectBean();
    sub.setMatricNo(matricNo);
    sub.setSubjectCode(request.getParameter("subjectCode"));
    sub.setSubjectName(request.getParameter("subjectName"));

    subjectDAO.addSubject(sub);
    response.sendRedirect("SubjectServlet?action=view");
}

// ===== 3. EDIT =====
else if ("edit".equals(action)) {
    int id = Integer.parseInt(request.getParameter("id"));
    SubjectBean existingSub = subjectDAO.getSubjectById(id);
    request.setAttribute("subject", existingSub);
    request.getRequestDispatcher("subject/updateSubject.jsp").forward(request, response);
}

// ===== 4. UPDATE =====
else if ("update".equals(action)) {
    SubjectBean sub = new SubjectBean();
    sub.setId(Integer.parseInt(request.getParameter("id")));
    sub.setMatricNo(matricNo);
    sub.setSubjectCode(request.getParameter("subjectCode"));
    sub.setSubjectName(request.getParameter("subjectName"));

    subjectDAO.updateSubject(sub);
    response.sendRedirect("SubjectServlet?action=view");
}

// ===== 5. DELETE =====
else if ("delete".equals(action)) {
    int id = Integer.parseInt(request.getParameter("id"));
    subjectDAO.deleteSubject(id, matricNo);
    response.sendRedirect("SubjectServlet?action=view");
}
}
}

```

SubjectServlet

```
package com.lab.bean;

public class SubjectBean {
    private int id;
    private String matricNo;
    private String subjectCode;
    private String subjectName;

    public SubjectBean() {}

    // Getters and Setters
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getMatricNo() { return matricNo; }
    public void setMatricNo(String matricNo) { this.matricNo = matricNo; }

    public String getSubjectCode() { return subjectCode; }
    public void setSubjectCode(String subjectCode) { this.subjectCode = subjectCode; }

    public String getSubjectName() { return subjectName; }
    public void setSubjectName(String subjectName) { this.subjectName = subjectName; }
}
```

SubjectBean

```

package com.iab.dao;

import com.iab.bean.SubjectBean;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.ArrayList;

public class SubjectDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db", "root", "admin");
    }

    // 1. CREATE
    public boolean addSubject(SubjectBean subject) {
        String sql = "INSERT INTO registered_subjects (matric_no, subject_code, subject_name) VALUES (?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, subject.getMatricNo());
            pstmt.setString(2, subject.getSubjectCode());
            pstmt.setString(3, subject.getSubjectName());
            return pstmt.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }

    // 2. READ
    public ArrayList<SubjectBean> getSubjectsByStudent(String matricNo) {
        ArrayList<SubjectBean> list = new ArrayList<>();
        String sql = "SELECT * FROM registered_subjects WHERE matric_no = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, matricNo);
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                SubjectBean sub = new SubjectBean();
                sub.setId(rs.getInt("id"));
                sub.setMatricNo(rs.getString("matric_no"));
                sub.setSubjectCode(rs.getString("subject_code"));
                sub.setSubjectName(rs.getString("subject_name"));
                list.add(sub);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```



```

// 3.
public SubjectBean getSubjectById(int id) {
    String sql = "SELECT * FROM registered_subjects WHERE id = ?";
    try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setInt(1, id);
        ResultSet rs = pstmt.executeQuery();
        if (rs.next()) {
            SubjectBean sub = new SubjectBean();
            sub.setId(rs.getInt("id"));
            sub.setMatricNo(rs.getString("matric_no"));
            sub.setSubjectCode(rs.getString("subject_code"));
            sub.setSubjectName(rs.getString("subject_name"));
            return sub;
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return null;
}

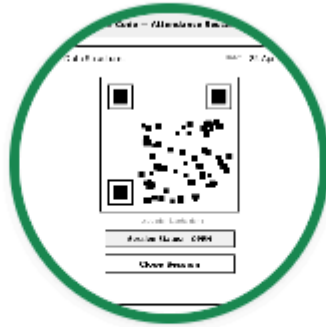
// 4. UPDATE
public boolean updateSubject(SubjectBean subject) {
    String sql = "UPDATE registered_subjects SET subject_code = ?, subject_name = ? WHERE id = ? AND matric_no = ?";
    try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setString(1, subject.getSubjectCode());
        pstmt.setString(2, subject.getSubjectName());
        pstmt.setInt(3, subject.getId());
        pstmt.setString(4, subject.getMatricNo());
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

// 5. DELETE
public boolean deleteSubject(int id, String matricNo) {
    String sql = "DELETE FROM registered_subjects WHERE id = ? AND matric_no = ?";
    try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setInt(1, id);
        pstmt.setString(2, matricNo);
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}
}

```

subjectDao

OUTPUT



Welcome, 1!

Matric Number: 1

Subject Management

[View My Registered Subjects](#)

[Register New Subject](#)

[Logout](#)

[Delete Account](#)

Register New Subject

Subject Code:

Subject Name:

[Cancel and Back](#)

Registered Academic Subjects

[+ Register New Subject](#) | [Dashboard](#)

No	Subject Code	Subject Name	Actions
1	123	web app	Edit Delete Register New Subject Back to Dashboard
2	789	APA SUBJECT	Edit Delete Register New Subject Back to Dashboard

Edit Subject Details

Subject Code:

Subject Name:

[Cancel](#)

Registered Academic Subjects

[+ Register New Subject](#) | [Dashboard](#)

No	Subject Code	Subject Name	Actions
1	123	web app	Edit Delete Register New Subject Back to Dashboard
2	789	APA SUBJECT??	Edit Delete Register New Subject Back to Dashboard

Registered Academic Subjects

[+ Register New Subject](#) | [Dashboard](#)

No	Subject Code	Subject Name	Actions
1	123	web app	Edit Delete Register New Subject Back to Dashboard
2	789	APA SUBJECT??	Edit Delete Register New Subject Back to Dashboard

localhost:8080 says

Adakah anda pasti mahu memadam subjek ini? (Are you sure you want to delete this subject?)

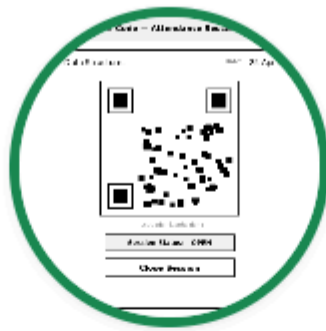
OK

Cancel

Registered Academic Subjects

[+ Register New Subject](#) | [🏠 Dashboard](#)

No	Subject Code	Subject Name	Actions
1	123	web app	Edit Delete Register New Subject Back to Dashboard



Welcome, 1!

Matric Number: 1

Subject Management

View My Registered Subjects

Register New Subject

Logout

Delete Account

Registered Academic Subjects

[+ Register New Subject](#) | [Dashboard](#)

No	Subject Code	Subject Name	Actions
1	123	web app	Edit Delete Register New Subject Back to Dashboard